IOWA STATE UNIVERSITY

LAB 3 - COMPOSITE BEAM REPORT

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Aer E 4260 Design of Aerospace Structures

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Contents

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Introduction

In lab 3, we are tasked with designing a composite beam to support a 350 lb camera, which is positioned at the free end of the beam. The primary constraint in this design is minimizing deflection, such that the beam does not move more than 1/50 of an inch under the load. The beam has a fixed-supported end and a free end where the load is applied.

To meet this deflection constraint, we will calculate the minimum moment of inertia required and evaluate two different composite beam configurations: (1) a quasi-isotropic laminate and (2) a custom design with a higher modulus than the quasi-isotropic laminate. Each lamina is 0.002 inches thick, and the properties of the composite material are given, including a density of 0.057 lb/in^3, and tables with Young's modulus, shear modulus, ultimate axial strength in tension, ultimate axial strength in compression, and ultimate shea strength.

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Tables

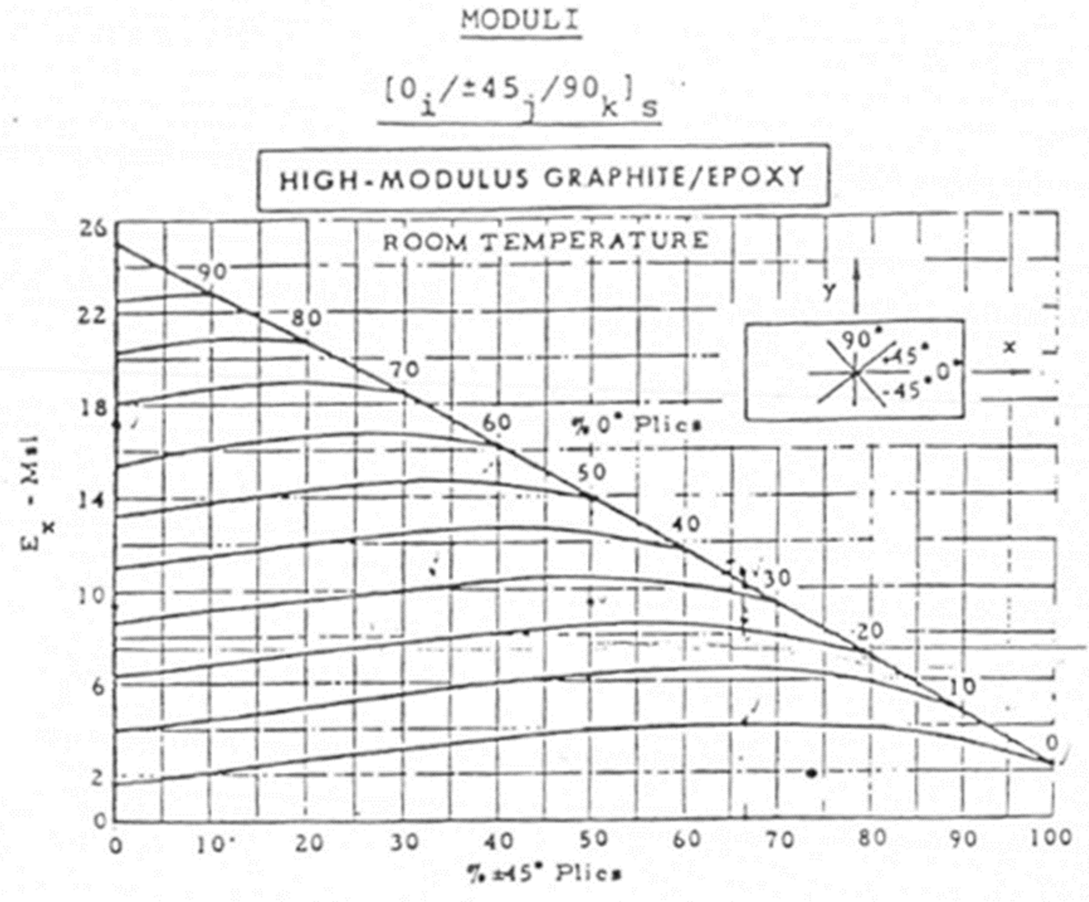


Table 2.1: *Young's modulus E (Msi) by plies %*

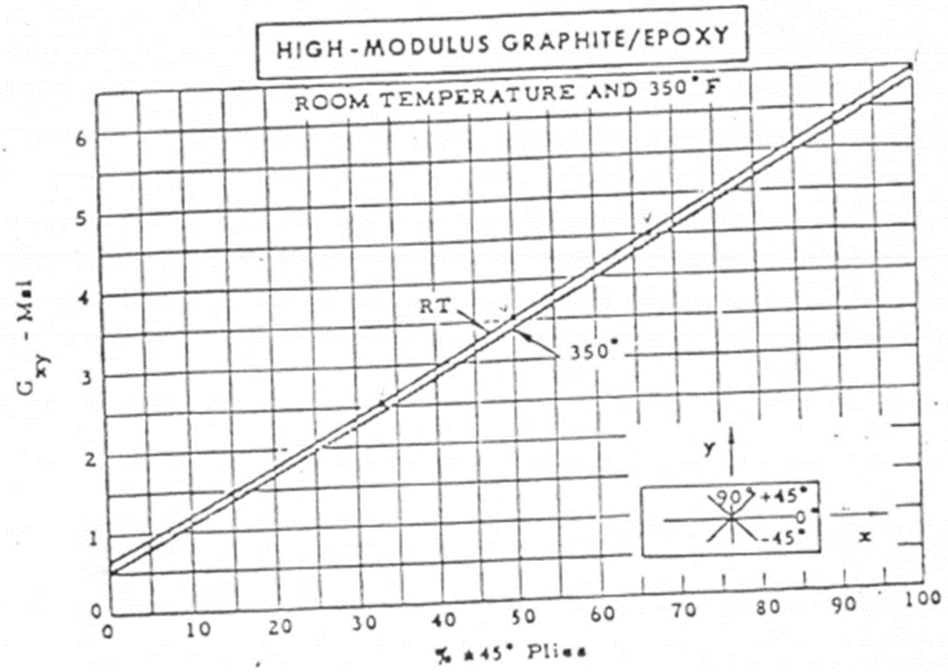


Table 2.2: *Shear modulus G (Msi) by plies %*

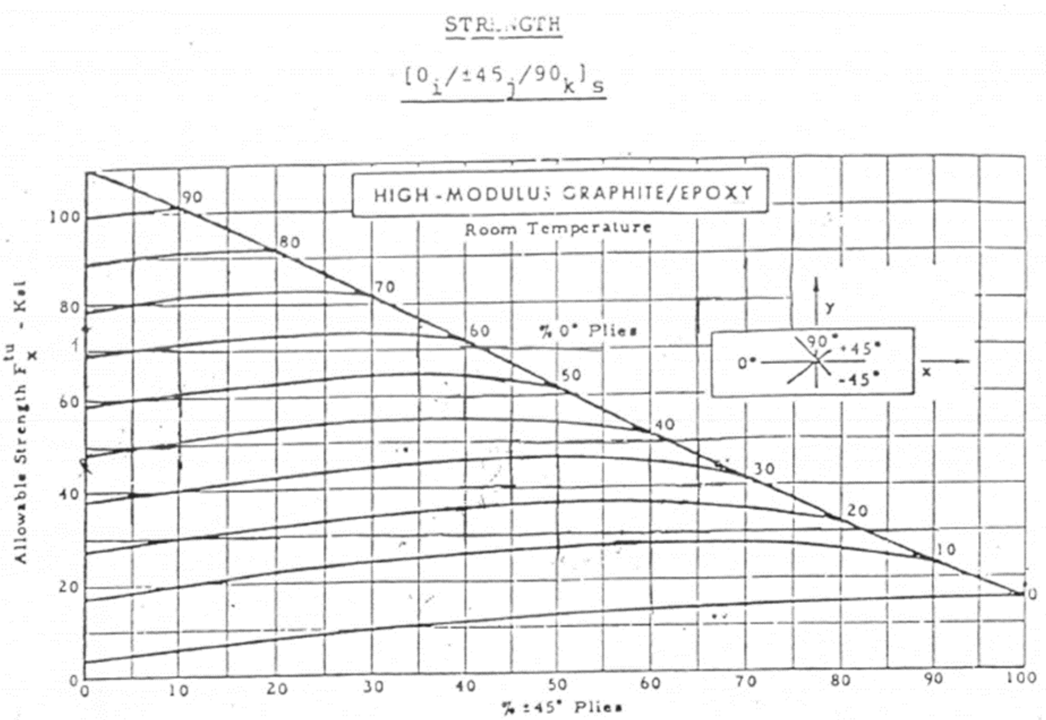


Table 2.3: *Ultimate axial strength in tension $\sigma_{T,u}$ (ksi) by plies %*

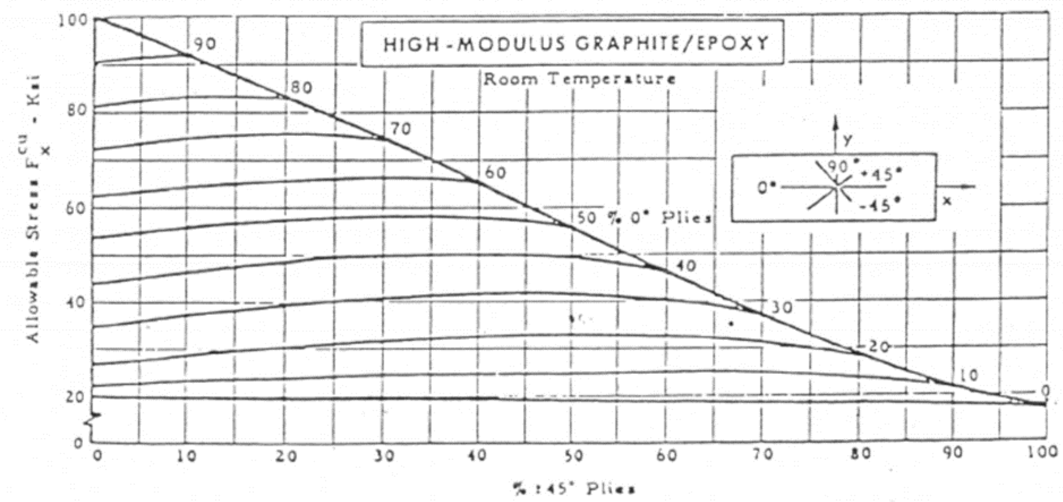


Table 2.4: *Ultimate axial strength in compression $\sigma_{C,u}$ (ksi) by plies %*

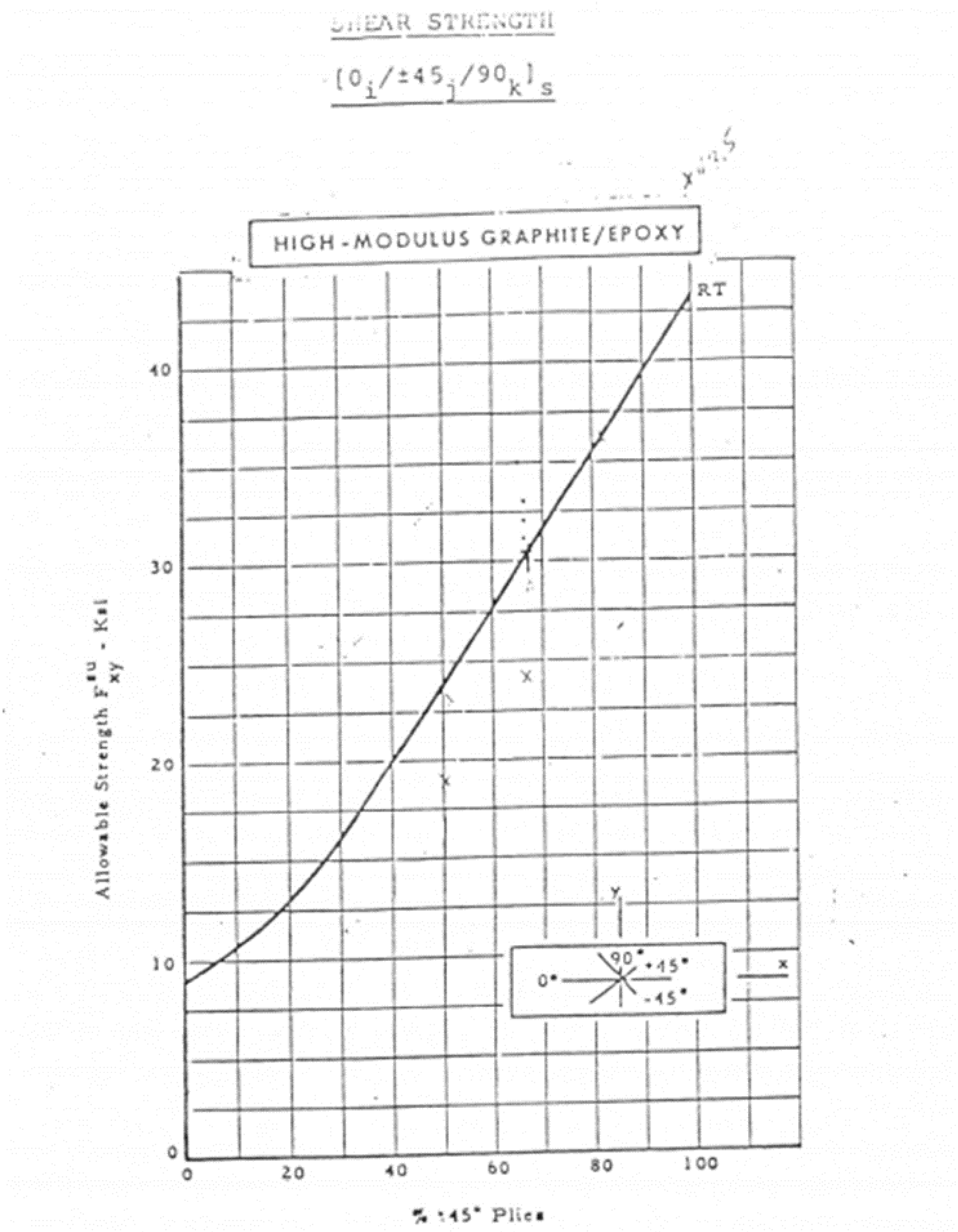


Table 2.5: *Ultimate shear strength τ_u (ksi) by plies %*

	Item#	d-Depth	t-Web	b-Flange	t-flange	A	I	Q	Weight	I/c	It/Q
		inch	inch	inch	inch	inch^2	inch^4		lbs/ft	in^3	
1	IB317061	3	0.17	2.33	0.17	1.3	1.85	0.7854	1.96	1.2333	0.4004
2	IB334961	3	0.349	2.509	0.349	2.8	3.45	1.7061	2.59	2.3000	0.7057
3	IB419061	4	0.19	2.66	0.19	1.77	4.42	1.3908	2.65	2.2100	0.6038
4	IB432661	4	0.326	2.796	0.326	3.13	7.19	2.4750	3.28	3.5950	0.9470
5	IB521061	5	0.21	3	0.21	2.31	8.91	2.2313	3.43	3.5640	0.8386
6	IB549461	5	0.494	3.284	0.494	5.71	19.19	5.5995	5.1	7.6760	1.6930
7	IB623061	6	0.23	3.33	0.23	2.91	16.02	3.3327	4.3	5.3400	1.1056
8	IB634361	6	0.343	3.443	0.343	4.42	23.21	5.0863	5.1	7.7367	1.5652
9	IB827061	8	0.27	4	0.27	4.32	41.62	6.4800	6.35	10.4050	1.7342
10	IB1031061	10	0.31	4.66	0.31	5.99	89.16	11.0980	8.76	17.8320	2.4905
11	IB1235061	12	0.35	5	0.35	7.7	160.88	16.8000	11	26.8133	3.3517
12	IB313061	3	0.15	2.5	0.26	1.75	2.64	1.1438	2.03	1.7600	0.3462
13	IB315061	4	0.17	3	0.29	2.42	6.57	2.0800	2.79	3.2850	0.5370
14	IB417061	5	0.19	3.5	0.32	3.19	13.6	3.3938	3.7	5.4400	0.7614
15	IB519061	6	0.19	4	0.29	3.46	21.45	4.3350	4.03	7.1500	0.9401
16	IB619061	6	0.21	4	0.35	4.06	24.98	5.1450	4.69	8.3267	1.0196
17	IB621061	7	0.21	3.5	0.38	4.13	33.43	5.9413	4.72	9.5514	1.1816
18	IB721061	7	0.23	4.5	0.38	5.03	42.17	7.3938	5.8	12.0486	1.3118
19	IB723061	8	0.23	5	0.35	5.34	58.7	8.8400	6.18	14.6750	1.5273
20	IB823061	8	0.25	5	0.41	6.1	66.82	10.2000	7.02	16.7050	1.6377
21	IB825061	10	0.25	6	0.41	7.42	129.31	15.4250	8.65	25.8620	2.0958
22	IB1025061	12	0.31	7	0.62	12.4	313.48	31.6200	14.29	52.2467	3.0733
23	IB1231061	3	0.13	2.5	0.2	1.39	2.15	0.8963	1.64	1.4333	0.3119
24	WF44313	4	0.313	4	0.313	3.76	13.47	3.1300	4.76	6.7350	1.3470
25	WF55313	5	0.313	5	0.313	4.7	35.95	4.8906	6.44	14.3800	2.3008

Table 2.6: *Aluminum I-beams and their dimensions and properties*

3

Methodology

3.1 Quasi-Isotropic Laminate Beam

Starting with the quasi-isotropic laminate ($n_1 = n_2 = n_3$), the first step was to address the primary design constraint: maximum displacement $\delta_{\max} = \frac{1}{50}$ in = 0.02 in. To do so, we can calculate the minimum moment of inertia using the formula (3.1), assuming displacement due to shear is neglected. From Table 2.1, Young's modulus $E = 9.5$ Msi for a quasi-isotropic laminate.

$$I_{\min} = \frac{PL^3}{3E\delta_{\max}} = \frac{(0.35 \text{ kip})(60 \text{ in})^3}{3(9.5 \times 10^3 \text{ ksi})(0.02 \text{ in})} = 132.632 \text{ in}^4 \tag{3.1}$$

With the known $I_{\min}$ value, from Table 2.6, I chose an aluminum beam with a close I value, IB825061 with $I = 129.31$ in^4. Its dimensions are $d = 10$ in, $b = 6$ in, $t_w = 0.25$ in, and $t_f = 0.41$ in.

For this beam's thicknesses, we would have:

$$\frac{0.25 \text{ in}}{0.002 \text{ in/laminate}} = 125 \text{ laminates (web)}$$
$$\frac{0.41 \text{ in}}{0.002 \text{ in/laminate}} = 205 \text{ laminates (flange)}$$

Since the composite is symmetric, the final thicknesses were adjusted to $t_w = 0.252$ in and $t_f = 0.412$ in. Thus, the stacking sequence of the web was $[0_{16}/{+45}_{16}/{-45}_{16}/90_{15}]_s$ (total of 126 laminates), and for the flange was $[0_{26}/{+45}_{26}/{-45}_{26}/90_{25}]_s$ (total of 206 laminates). The depth and width were kept the same, $d = 10$ in and $b = 6$ in. With those values, the area and the actual moment of inertia were calculated as follows:

$$A = 7.464 \text{ in}^2 \tag{3.2}$$

$$I = \frac{b(d + 2t_f)^3 - (b - t_w)d^3}{12} \tag{3.3}$$

$$I = \frac{(6 \text{ in})(10 \text{ in} + 2 \times 0.412 \text{ in})^3 - (6 \text{ in} - 0.252 \text{ in})(10 \text{ in})^3}{12} = 155.064 \text{ in}^4 \tag{3.4}$$

From (Table 2.2), Shear modulus $G = 3.5$ Msi for a quasi-isotropic laminate. With those values, the actual displacement at the free end of the beam was calculated as:

$$\delta = \frac{PL^3}{3EI} + \frac{PL}{GA} \tag{3.5}$$

$$\delta = \frac{(0.35\text{ kip})(60\text{ in})^3}{3(9.5\times10^3\text{ ksi})(155.064\text{ in}^4)} + \frac{(0.35\text{ kip})(60\text{ in})}{(3.5\times10^3\text{ ksi})(7.464\text{ in}^2)} = 0.0179\text{ in} \tag{3.6}$$

Thus, we confirm that we are within the allowable displacement of $\frac{1}{50}$ in.

To calculate the factors of safety, first, the maximum axial stress and shear stress were calculated:

$$M = PL = (0.35\text{ kip})(60\text{ in}) = 21\text{ kip-in} \tag{3.7}$$

$$\sigma_{\max} = \frac{Mc}{I} = \frac{(21\text{ kip-in})(10/2 + 0.412\text{ in})}{155.064\text{ in}^4} = 7.545\text{ ksi} \tag{3.8}$$

$$V = P = 0.35\text{ kip} \tag{3.9}$$

$$Q = bt_f\frac{(d+2t_f)}{2} + t_w\frac{(d+2t_f)^2}{8} \tag{3.10}$$

$$Q = (6\text{ in})(0.412\text{ in})\frac{(10\text{ in} + 2\times0.412\text{ in})}{2} + (0.252\text{ in})\frac{(10\text{ in} + 2\times0.412\text{ in})^2}{8} = 17.069\text{ in}^3 \tag{3.11}$$

$$\tau_{\max} = \frac{VQ}{It_w} = \frac{(0.35\text{ kip})(17.069\text{ in}^3)}{(155.064\text{ in}^4)(0.252\text{ in})} = 0.153\text{ ksi} \tag{3.12}$$

Now, we calculate the factors of safety in displacement, tension, compression, and shear, with the ultimate strengths retrieved from (Table 2.3), (Table 2.4), and (Table 2.5):

$$FS_\delta = \frac{\delta_{\max}}{\delta} = \frac{0.02\text{ in}}{0.0179\text{ in}} = 1.12 \tag{3.13}$$

$$FS_{\sigma_T} = \frac{\sigma_{T,u}}{\sigma_{\max}} = \frac{43\text{ ksi}}{7.545\text{ ksi}} = 5.70 \tag{3.14}$$

$$FS_{\sigma_C} = \frac{\sigma_{C,u}}{\sigma_{\max}} = \frac{37\text{ ksi}}{7.545\text{ ksi}} = 4.90 \tag{3.15}$$

$$FS_\tau = \frac{\tau_u}{\tau_{\max}} = \frac{23\text{ ksi}}{0.153\text{ ksi}} = 150 \tag{3.16}$$

Lastly, the beam weight was calculated as:

$$W = AL\rho = (7.464\text{ in}^2)(60\text{ in})(0.057\text{ lb/in}^3) = 25.5\text{ lb} \tag{3.17}$$

3.2 My Beam Design

For my own laminate design, I chose the laminate configuration $n_1 = 80\%$, $n_2 = 10\%$, and $n_3 = 10\%$. For this configuration, I retrieved the following values from the tables:

$$\begin{aligned} E &= 21 \text{ Msi} \\ G &= 1.3 \text{ Msi} \\ \sigma_{T,u} &= 92 \text{ ksi} \\ \sigma_{C,u} &= 83 \text{ ksi} \\ \tau_u &= 11 \text{ ksi} \end{aligned}$$

Calculating I_{min}:

$$I_{\text{min}} = \frac{(0.35 \text{ kip})(60 \text{ in})^3}{3(2.1 \times 10^4 \text{ ksi})(0.02 \text{ in})} = 60 \text{ in}^4 \tag{3.18}$$

From I_{min}, I chose the aluminum beam IB723061 from (Table 2.6) as a reference. Its dimensions are $t_w = 0.23$ in and $t_f = 0.35$ in. These were adjusted to $t_w = 0.232$ in and $t_f = 0.352$ in for even laminates. Thus, the stacking sequence of the web was $[0_{46}/{+45_3}/{-45_3}/90_6]_s$ (total of 116 laminates), and for the flange was $[0_{71}/{+45_4}/{-45_4}/90_9]_s$ (total of 176 laminates). The depth $d = 8$ in and width $b = 5$ in. With the dimensions defined, the area, actual moment of inertia, and displacement were calculated:

$$A = 5.376 \text{ in}^2 \tag{3.19}$$

$$I = \frac{(5 \text{ in})(8 \text{ in} + 2 \times 0.352 \text{ in})^3 - (5 \text{ in} - 0.232 \text{ in})(8 \text{ in})^3}{12} = 71.320 \text{ in}^4 \tag{3.20}$$

$$\delta = \frac{(0.35 \text{ kip})(60 \text{ in}^3)}{3(2.1 \times 10^4 \text{ ksi})(71.320 \text{ in}^4)} + \frac{(0.35 \text{ kip})(60 \text{ in})}{(1.3 \times 10^3 \text{ ksi})(5.376 \text{ in}^2)} = 0.0198 \text{ in} \tag{3.21}$$

Once again, the displacement is lower than the maximum allowable displacement of $\frac{1}{50}$ in.

The factors of safety and weight were calculated as:

$$\sigma_{\text{max}} = \frac{(21 \text{ kip-in})(8/2 + 0.352 \text{ in})}{71.320 \text{ in}^4} = 1.281 \text{ ksi} \tag{3.22}$$

$$Q = (5 \text{ in})(0.352 \text{ in})\frac{(8 \text{ in} + 2 \times 0.352 \text{ in})}{2} + (0.232 \text{ in})\frac{(8 \text{ in} + 2 \times 0.352 \text{ in})^2}{8} = 9.857 \text{ in}^3 \tag{3.23}$$

$$\tau_{\max} = \frac{(0.35\text{ kip})(9.857\text{ in}^3)}{(71.320\text{ in}^4)(0.232\text{ in})} = 0.208\text{ ksi} \tag{3.24}$$

$$FS_{\delta} = \frac{0.02\text{ in}}{0.0198\text{ in}} = 1.01 \tag{3.25}$$

$$FS_{\sigma_T} = \frac{92\text{ ksi}}{1.281\text{ ksi}} = 71.8 \tag{3.26}$$

$$FS_{\sigma_C} = \frac{83\text{ ksi}}{1.281\text{ ksi}} = 64.8 \tag{3.27}$$

$$FS_{\tau} = \frac{11\text{ ksi}}{0.208\text{ ksi}} = 52.9 \tag{3.28}$$

$$W = (5.376\text{ in}^2)(60\text{ in})(0.057\text{ lb/in}^3) = 18.4\text{ lb} \tag{3.29}$$

4

Conclusion

In this analysis, we designed two 60-inch-long I-beams made of composite materials: one using a quasi-isotropic laminate configuration, and the other with a custom configuration of n1 = 80%, n2 = 10%, and n3 = 10%. Both beams met the maximum allowable displacement requirement and demonstrated high factors of safety. However, the custom laminate configuration exhibited significantly higher ultimate axial strengths and Young's modulus, although its ultimate shear strength and shear modulus were lower compared to the quasi-isotropic design. The custom design also had notably higher factors of safety overall.

For the custom laminate beam, the web consisted of 116 laminates, while each flange had 176 laminates. In terms of weight, the quasi-isotropic beam weighed 25.5 lb, whereas the custom laminate beam was lighter, weighing just 18.4 lb. Thus, the custom design not only provided better performance in terms of safety but also achieved a significant reduction in weight.

A

Appendix A

A.1 Rough Work

$\frac{1}{50}\ in = 0.02\ in$

AerE 426/526
Fall 2024
Lab Exercise -3

Due one week after the lab.

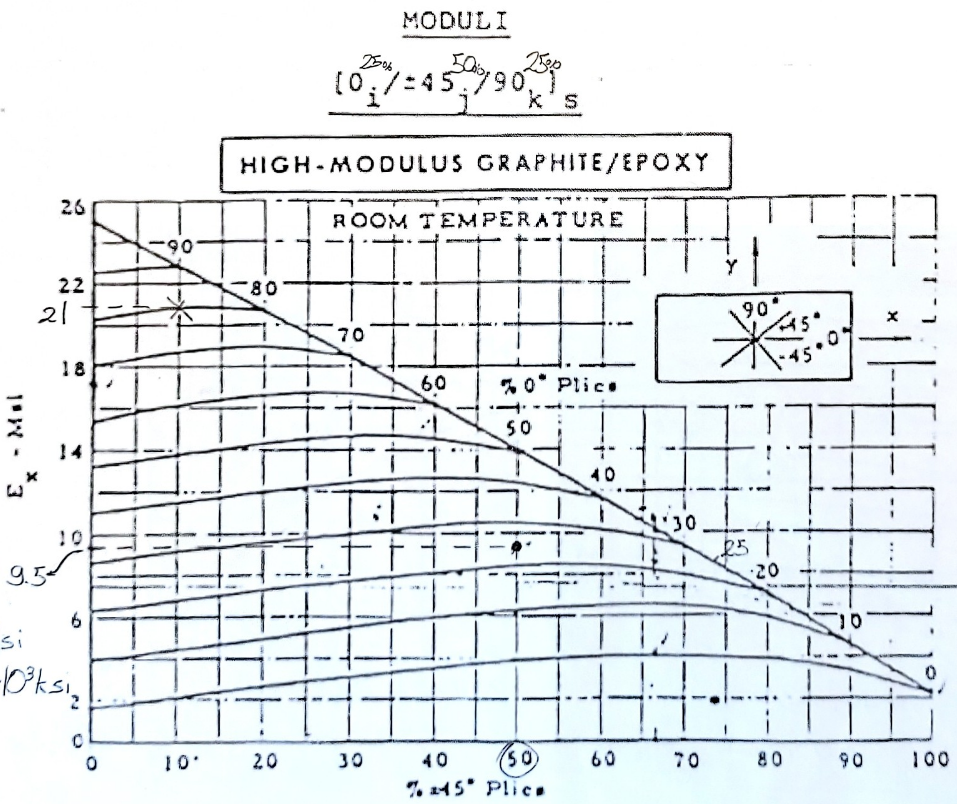
$P = 350\ lb = 0.35\ klb$

You have to design a 60-inch-long beam at the tip of it is a camera weighing 350 lb. To keep the distortion to the minimum, the beam should not move more than 1/50 of an inch when the camera is mounted. Assume the supported end is fixed and the camera end is free. Make it out of high-performance composite with properties charts given below and each lamina is 0.002 inch thick. Cost of the composite is \$5/lb. Density of composite 0.057 lb/in^3. Choose two different Composite moduli, (1) quasi-isotropic, and (2) your choice but make it higher than quasi-isotropic, perform the calculations outlined below for (1) and (2).

a. For the displacement, as a quick approximation, neglect the shear displacement to estimate the desired I. Since I does not depend on the material, you can choose the shape, based on the aluminum chart, included. Now based on composite requirements a laminate is ($[0_{n1} \therefore 45_{n2} \therefore -45_{n2} \therefore 90_{n3}]_s$) For quasi-isotropic laminate (1) use n1=n2=n3 and for your beam design (2) they will not be same. Calculate actual dimensions and get total displacement. Which beam will be lighter and by how much?
b. Based on your design and the thickness requirements, what stacking sequence of the laminas on the flange and web will be required?
c. Calculate the actual beam dimensions, Factor of safety in displacement, tension, compression, and shear.

Equations: $\sigma_{max} = \frac{Mc}{I}$; and $\tau_{max} = \frac{VQ}{It}$; I=(b*d^3-(b-tw)*(d-2tf)^3)/12; Q=b*tf*d/2+tw*d^2/8

$\delta = \delta_b + \delta_s = \frac{PL^3}{3EI} + \frac{PL}{GA}$ $\quad \delta_b = dispacement\ due\ to\ bending\ load, and\ \delta_s = dispacement\ due\ to\ shear$



(quasi-iso) 9.5

9.5 Msi $= 9.5\times10^3\ ksi$

$\cdot\ \delta = \delta_b + \cancel{\delta_s}$ neglected initially

$\delta = \delta_b = \frac{PL^3}{3EI}$

$\cdot\ I_{min} = \frac{PL^3}{3E\,\delta_{max}}$

$I_{min} = \frac{(0.35\ klb)(60\ in)^3}{3(9.5\cdot10^3\ ksi)(\frac{1}{50}\ in)}$

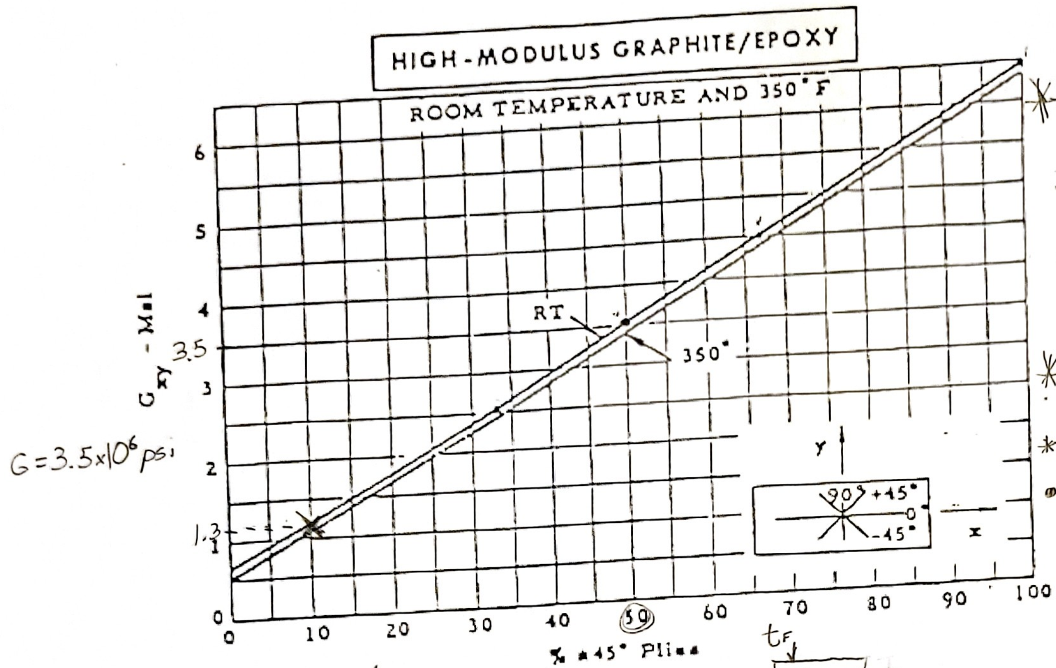
$I_{min} = 132.632\ in^4$

$\cdot$ Al beam chosen : 21

$\cdot\ t_{w,Al} = 0.25\ in$ $\quad$ * lamina thickness 0.002 in

$\frac{0.25\ in}{0.002\ in} = 125$ laminas

$\cdot$ We need even laminas (simetric layering)



$\Downarrow$

*. $t_w = 0.252$ (126 laminas)

$t_{F,re} = 0.41$ in

. $\frac{0.41\ in}{0.002\ in} = 205$ laminas

*. $t_F = 0.412$ in (206 laminas)

* Quasi-isotropic

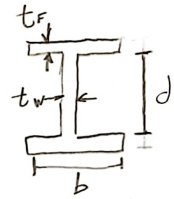
• Actual dimensions:

. $t_w = 0.252$ in

. $t_F = 0.412$ in

. $d = 10$ in

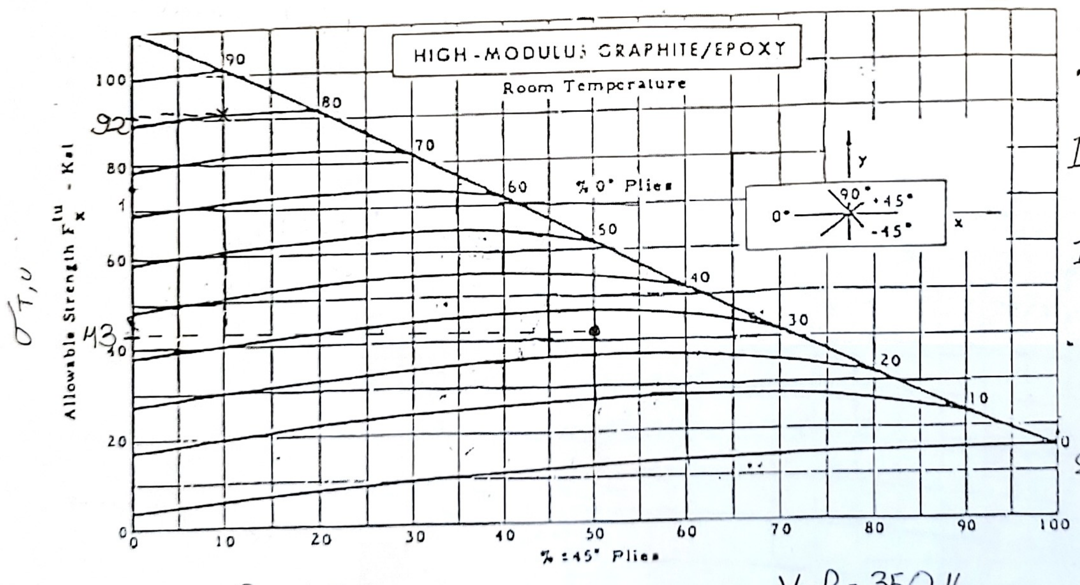
. $b = 6$ in



STRENGTH

$[0_i/\pm45_j/90_k]_S$

. $A = 7.464\ in^2$ (using online calculator)



. $I = \frac{b(d+2t_F)^3 - (b-t_w)d^3}{12}$

$I = \frac{6(10+2\cdot0.412)^3 - (6-0.252)\cdot10^3}{12}$

$I = 155.064\ in^4$

. $\delta = \delta_b + \delta_s$

$\delta = \frac{PL^3}{3EI} + \frac{PL}{GA}$

$\delta = \frac{(350\ lb)(60\ in)^3}{3(9.5\cdot10^6\ psi)(155.064\ in^4)} + \frac{(350\ lb)(60\ in)}{(3.5\times10^6\ psi)(7.464\ in^2)}$

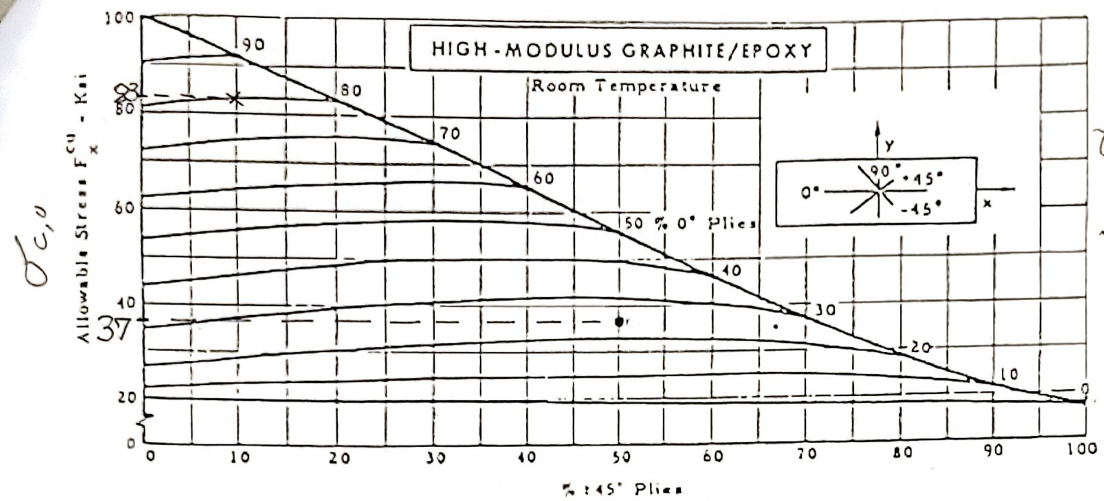
$\delta = 0.0179\ in < \frac{1}{50}\ in$ ✓

. $M = P\cdot L = 350\ lb\cdot 60\ in = 21\ kip\cdot in$

. $V = P = 350\ lb$

. $\sigma_{max} = \pm\frac{Mc}{I} = \frac{(21\ kip\cdot in)\left(\frac{10}{2}\ in + 0.412\ in\right)}{155.064\ in^4} = \pm 7.545\ ksi$

. $Q = b\cdot t_F\cdot\frac{(d+2t_F)}{2} + t_w\frac{(d+2t_F)^2}{8} = 6\cdot0.412\frac{(10+2\cdot0.412)}{2} + 0.252\frac{(10+2\cdot0.412)^2}{8} = 17.069\ in^3$



$$\tau_{max} = \frac{VQ}{I t_w}$$

$$\tau_{max} = \frac{(0.35\ kip)(17.069\ in^3)}{(155.064\ in^4)(0.252\ in)}$$

$$\tau_{max} = 0.153\ ksi$$

$$W = A \cdot L \cdot \rho$$

$$W = (7.464\ in^2)(60\ in)(0.057\ lb/in^3)$$

$$W = 25.5\ lb$$

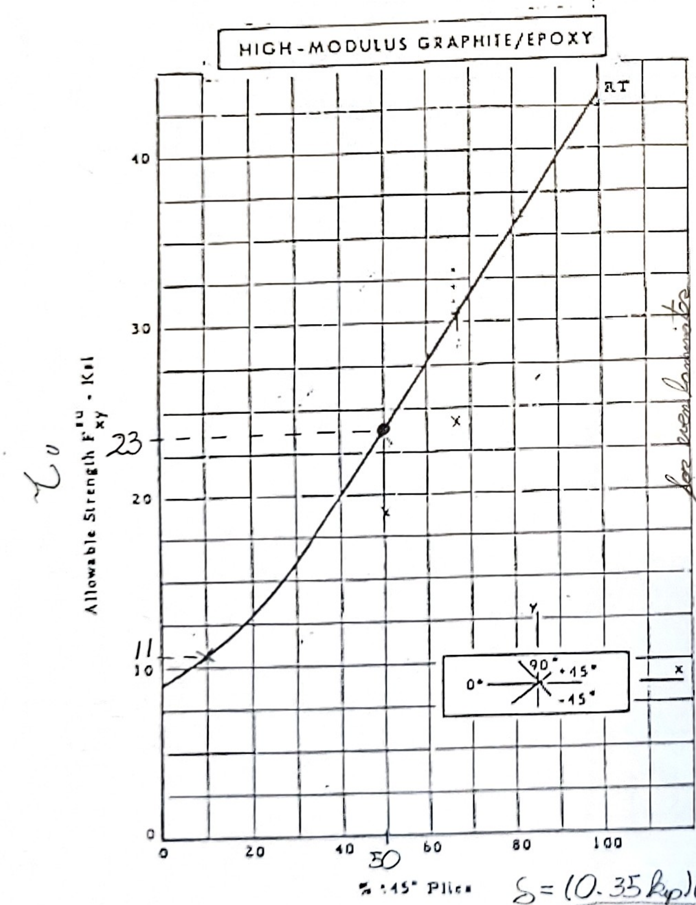
$$FS_\delta = \frac{\delta_{max}}{\delta} = \frac{0.02\ in}{0.0179\ in} = 1.12$$

SHEAR STRENGTH
$[0_i/\pm45_j/90_k]_s$

$$FS_{\sigma_T} = \frac{\sigma_{Tu}}{\sigma_{max}} = \frac{43\ ksi}{7.545\ ksi} = 5.70$$

$$FS_{\sigma_C} = \frac{\sigma_{cu}}{\sigma_{max}} = \frac{37\ ksi}{7.545\ ksi} = 4.90$$

$$FS_\tau = \frac{\tau_u}{\tau_{max}} = \frac{23\ ksi}{0.153\ ksi} = 150$$



My desing: $n_1 = 80\%$, $n_2 = 10\%$, $n_3 = 10\%$

- $E = 2.1 \times 10^4\ ksi$
- $I_{min} = \frac{(0.35\ kip)(60\ in)^3}{3(2.1\times10^4\ ksi)(1/50\ in)} = 60\ in^4$
- Al beam nº: 19 ($t_{w,Al} = 0.23\ in$; $t_{F,Al} = 0.35\ in$)

for beam laminates

(4) $t_w = 0.232\ in$ (116 laminates); $t_F = 0.352\ in$ (176 laminates)

- $d = 8\ in$; $b = 5\ in$
- $A = 5.376\ in^2$
- $I = \frac{5(8+2\cdot0.352)^3 - (5-0.232)8^3}{12}$

$$I = 71.320\ in^4$$

$$\delta = \frac{(0.35\ kip)(60\ in)^3}{3(2.1\times10^4\ ksi)(71.320\ in^4)} + \frac{(0.35\ kip)(60\ in)}{(1.3\times10^3\ ksi)(5.376\ in^2)}$$

$$\delta = 0.0198\ in < 0.02\ in\ \checkmark$$

	Item#	d-Depth	t-Web	b-Flange	t-flange	A	I	Q	Weight	I/c	It/Q
		inch	inch	inch	inch	inch^2	inch^4		lbs/ft	in^3	
1	IB317061	3	0.17	2.33	0.17	1.3	1.85	0.7854	1.96	1.2333	0.4004
2	IB334961	3	0.349	2.509	0.349	2.8	3.45	1.7061	2.59	2.3000	0.7057
3	IB419061	4	0.19	2.66	0.19	1.77	4.42	1.3908	2.65	2.2100	0.6038
4	IB432661	4	0.326	2.796	0.326	3.13	7.19	2.4750	3.28	3.5950	0.9470
5	IB521061	5	0.21	3	0.21	2.31	8.91	2.2313	3.43	3.5640	0.8386
6	IB549461	5	0.494	3.284	0.494	5.71	19.19	5.5995	5.1	7.6760	1.6930
7	IB623061	6	0.23	3.33	0.23	2.91	16.02	3.3327	4.3	5.3400	1.1056
8	IB634361	6	0.343	3.443	0.343	4.42	23.21	5.0863	5.1	7.7367	1.5652
9	IB827061	8	0.27	4	0.27	4.32	41.62	6.4800	6.35	10.4050	1.7342
10	IB1031061	10	0.31	4.66	0.31	5.99	89.16	11.0980	8.76	17.8320	2.4905
11	IB1235061	12	0.35	5	0.35	7.7	160.88	16.8000	11	26.8133	3.3517
12	IB313061	3	0.15	2.5	0.26	1.75	2.64	1.1438	2.03	1.7600	0.3462
13	IB315061	4	0.17	3	0.29	2.42	6.57	2.0800	2.79	3.2850	0.5370
14	IB417061	5	0.19	3.5	0.32	3.19	13.6	3.3938	3.7	5.4400	0.7614
15	IB519061	6	0.19	4	0.29	3.46	21.45	4.3350	4.03	7.1500	0.9401
16	IB619061	6	0.21	4	0.35	4.06	24.98	5.1450	4.69	8.3267	1.0196
17	IB621061	7	0.21	3.5	0.38	4.13	33.43	5.9413	4.72	9.5514	1.1816
18	IB721061	7	0.23	4.5	0.38	5.03	42.17	7.3938	5.8	12.0486	1.3118
19	IB723061	8	0.23	5	0.35	5.34	58.7	8.8400	6.18	14.6750	1.5273
20	IB823061	8	0.25	5	0.41	6.1	66.82	10.2000	7.02	16.7050	1.6377
21	IB825061	10	0.25	6	0.41	7.42	129.31	15.4250	8.65	25.8620	2.0958
22	IB1025061	12	0.31	7	0.62	12.4	313.48	31.6200	14.29	52.2467	3.0733
23	IB1231061	3	0.13	2.5	0.2	1.39	2.15	0.8963	1.64	1.4333	0.3119
24	WF44313	4	0.313	4	0.313	3.76	13.47	3.1300	4.76	6.7350	1.3470
25	WF55313	5	0.313	5	0.313	4.7	35.95	4.8906	6.44	14.3800	2.3008

$M = 21$ kip·in

$\sigma_{max} = \pm \dfrac{(21\,\text{kip·in})(\frac{8}{2} + 0.352)_{in}}{71.320\,in^4} = \pm 1.281$ ksi

$V = 0.35$ kip

$Q = \dfrac{5 \cdot 0.352(8+2 \cdot 0.352)}{2} + \dfrac{0.232(8+2 \cdot 0.352)^2}{8} = 9.857\,in^3$

$\tau_{max} = \dfrac{(0.35\,\text{kip})(9.857\,in^3)}{(71.320\,in^4)(0.232\,in)} = 0.208$ ksi

$W = (5.376\,in^2)(60\,in)(0.057\,lb/in^3)$

$W = 18.4\,lb$

$FS_{\delta} = \dfrac{0.02\,in}{0.0198\,in} = 1.01$

$FS_{\sigma_T} = \dfrac{92\,ksi}{1.281\,ksi} = 71.8$

$FS_{\sigma_C} = \dfrac{83\,ksi}{1.281\,ksi} = 64.8$

$FS_{\tau} = \dfrac{11\,ksi}{0.208\,ksi} = 52.9$